Univeristy of Pannonia

Specialist Course in Artificial intelligence solution development based on data- and systems science postgraduate specialization programme

Thesis report

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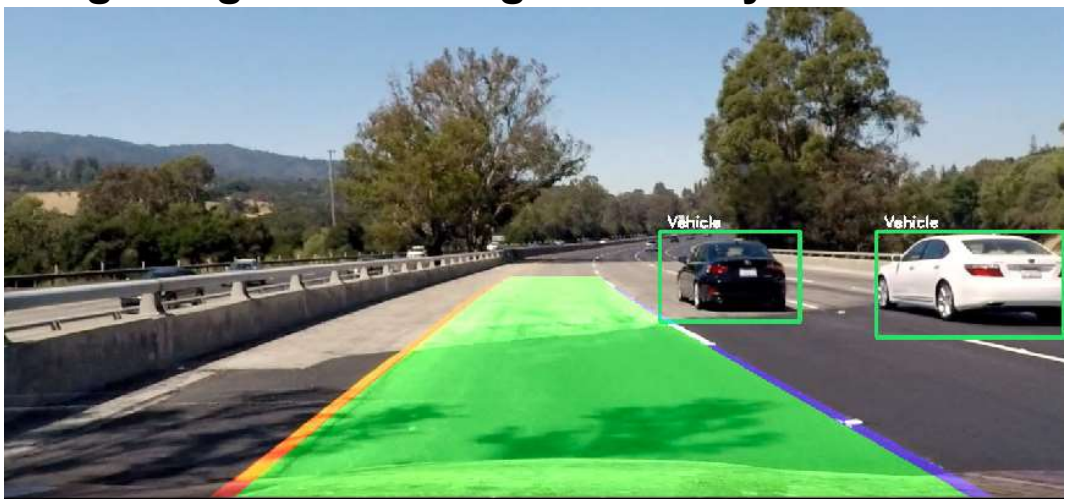
Lane Detection Using Deep Learning Methods

- **Introduction to the task**
- **Available methods**
- **State-of-the-art models**
- **Applied U-Net model overview**
- **Optimization and fine-tuning**
- **Extension with vehicle detection**

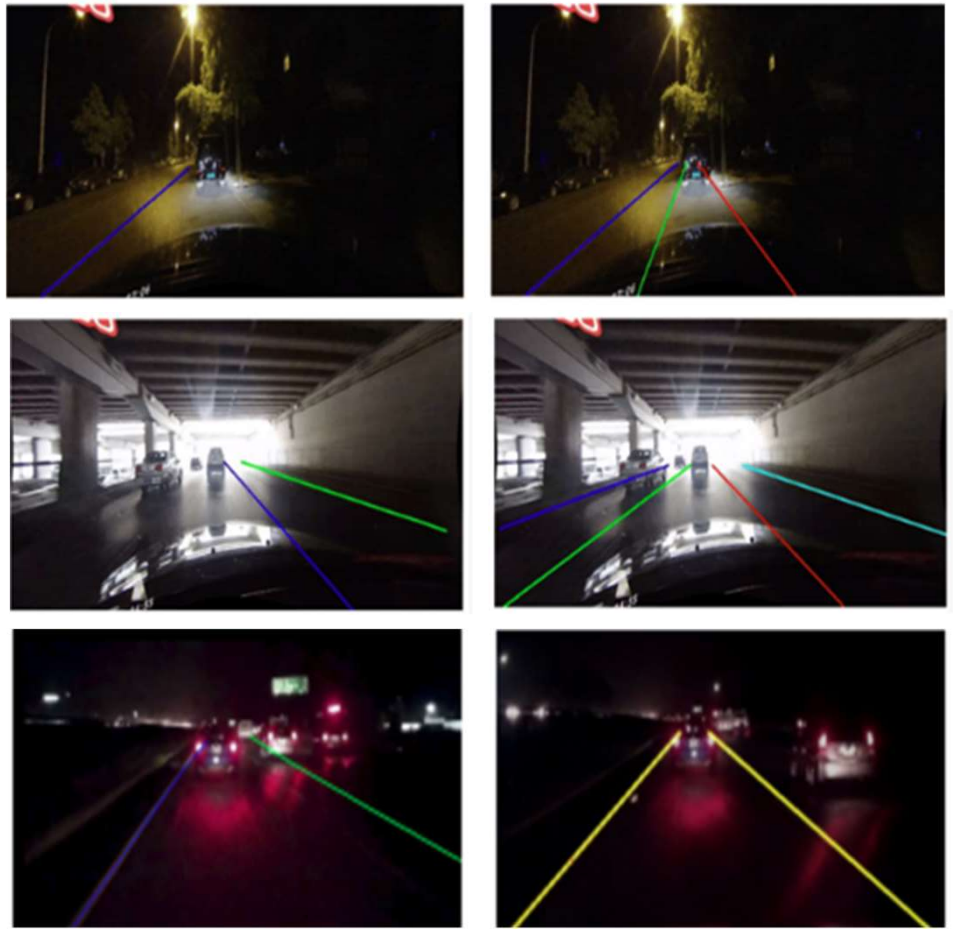
The task

- Investigation of lane detection methodologies
- Searching for datasets
- Implementing a lane detection method
- Model optimization
- Detecting moving vehicles → object detection
- Integrating and running the two systems



Challenges in Lane Detection

- **Changing weather conditions**
- **Varying traffic conditions**
- **Faulty road markings**



State of the art and Benchmarks

Best models depend on the dataset

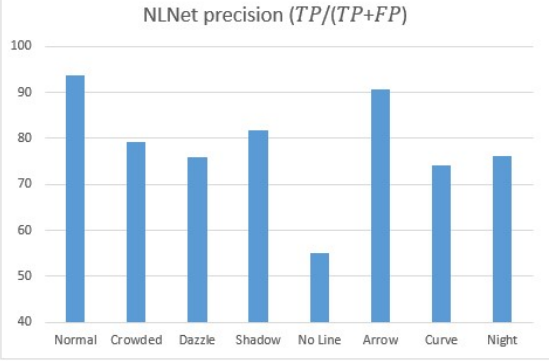
Dataset	Model	Method
TUSimple	RESA	Segmentation
CULane	CLRNet	"Anchor box"
CurveLanes	CondLaneNet	"Anchor box"
Llamas	CLRNet	"Anchor box"
BDD100k	YoloPv2	Segmentation

Environmental conditions strongly affect results

State of the art and Benchmarks

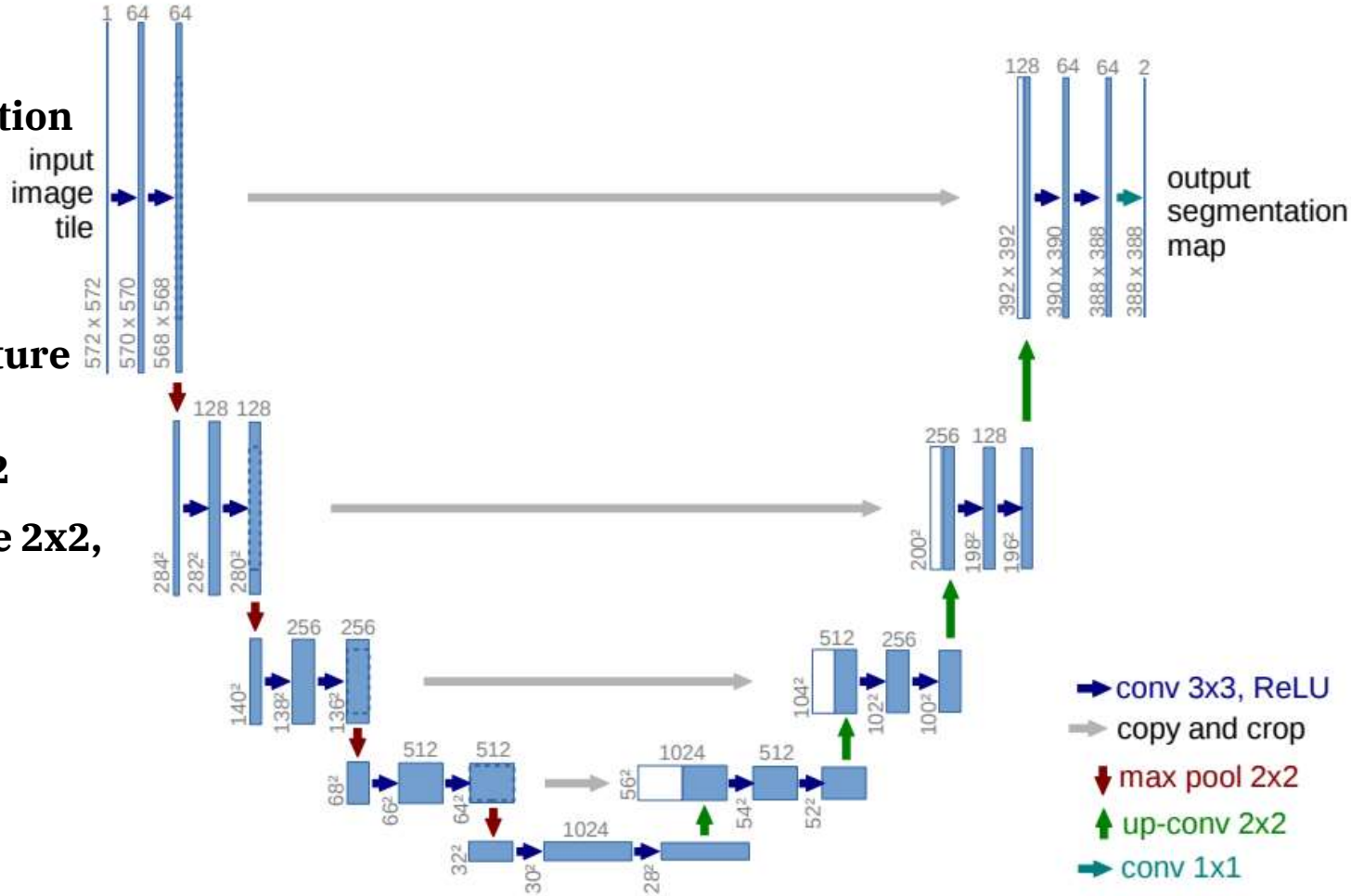
CULane Dataset

Method	Backbone	mF1	F1@50	F1@75	FPS	Normal	Crowded	Dazzle	Shadow	No Line	Arrow	Curve	Cross	Night
SCNN	VGG16	38.84	71.60	39.84	7.5	90.60	69.70	58.50	66.90	43.40	84.10	64.40	1990	66.10
RESA	ResNet34	-	74.50	-	45.5	91.90	72.40	66.50	72.00	46.30	88.10	68.60	1896	69.80
RESA	ResNet50	47.86	75.30	53.39	35.7	92.10	73.10	69.20	72.80	47.70	88.30	70.30	1503	69.90
FastDraw	ResNet50	-	-	-	90.3	85.90	63.60	57.00	69.90	40.60	79.40	65.20	7013	57.80
E2E	ERFNet	-	74.00		-	91.00	73.10	64.50	74.10	46.60	85.80	71.90	2022	67.90
UFLD	ResNet18	38.94	68.40	40.01	282	87.70	66.00	58.40	62.80	40.20	81.00	57.90	1743	62.10
UFLD	ResNet34		72.30	-	170	90.70	70.20	59.50	69.30	44.40	85.70	69.50	2037	66.70
PINet	Hourglass	46.81	74.40	51.33	25	90.30	72.30	66.30	68.40	49.80	83.70	65.20	1427	67.70
LaneATT	ResNet18	47.35	75.13	51.29	153	91.17	72.71	65.82	68.03	49.13	87.82	63.75	**1020**	68.58
LaneATT	ResNet34	49.57	76.68	54.34	129	92.14	75.03	66.47	78.15	49.39	88.38	67.72	1330	70.72
LaneATT	ResNet122	51.48	77.02	57.5	20	91.74	76.16	69.47	76.31	50.46	86.29	64.05	1264	70.81
LaneAF	ERFNet	48.6	75.63	54.53	24	91.10	73.32	69.71	75.81	50.62	86.86	65.02	1844	70.90
LaneAF	DLA34	50.42	77.41	56.79	20	91.80	75.61	71.78	79.12	51.38	86.88	72.70	1360	73.03
SGNet	ResNet18	-	76.12	-	117	91.42	74.05	66.89	72.17	50.16	87.13	67.02	1164	70.67
SGNet	ResNet34	-	77.27	-	92	92.07	75.41	67.75	74.31	50.90	87.97	69.65	1373	72.69
FOLOLane	ERFNet	-	78.80	-	40	92.70	77.80	75.20	79.30	52.10	89.00	69.40	1569	74.50
CondLane	ResNet18	51.84	78.14	57.42	173	92.87	75.79	70.72	80.01	52.39	89.37	72.40	1364	73.23
CondLane	ResNet34	53.11	78.74	59.39	128	93.38	77.14	71.17	79.93	51.85	89.89	73.88	1387	73.92
CondLane	ResNet101	54.83	79.48	61.23	47	93.47	77.44	70.93	80.91	54.13	90.16	**75.21**	1201	74.80
CLRNet	DLA34	55.64	80.47	62.78	94	93.73	**79.59**	75.30	**82.51**	54.58	90.62	74.13	1155	75.37
NLNet(ours)	DLA34	**55.75**	**80.49**	**63.16**	83	**93.74**	79.19	**75.85**	81.83	**55.03**	**90.69**	74.14	1291	**76.17**



Experiment with segmentation networks: U-Net

- **Experiment with segmentation networks: U-Net**
- **Created by Ronneberger et al. in 2015**
- **Encoder–Decoder architecture**
- **Encoder: Conv2D 3x3, MaxPooling2D 2x2**
- **Decoder: Conv2DTranspose 2x2, Concatenate, Conv2D 3x3**
- **Skip connections**



TUSimple Dataset

- **6408 pcs highway pictures**
- **3626 pcs train + 2782 pcs test**
- **1280x720 resolution**
- **1 sec video -> 20 pcs pictures -> last one is annoted**
- **annotation: .json file ; the pixels with lanes parts**


```
{
  "lanes": [
        [-2, -2, -2, -2, 632, 625, 617, 609, 601, 594,
        [-2, -2, -2, -2, 719, 734, 748, 762, 777, 791,
        [-2, -2, -2, -2, -2, 532, 503, 474, 445, 416,
        [-2, -2, -2, 781, 822, 862, 903, 944, 984, 102
      ],
  "h_samples": [240, 250, 260, 270, 280, 290, 300, 310,
  "raw_file": "path_to_clip"
}
```

Applying U-net

- **Data: TUSimple dataset**
- **Input pic: 1280x720x3 -> 256x256x3**
- **Input mask: .json -> 256x256x1 (binary)**
- **Train: 1800 pcs pics**
- **Test: 320 pcs pics**

```
Total params: 1,941,105 (7.40 MB)
Trainable params: 1,941,105 (7.40 MB)
Non-trainable params: 0 (0.00 B)
```

```
with strategy.scope():
    model = create_unet_model()
    model.compile(optimizer='adamw', loss='dice', metrics=['accuracy'])

callbacks = [
        tf.keras.callbacks.CSVLogger('/kaggle/working/log_tuned_adamw_dice_batch16.csv', append=False),
        tf.keras.callbacks.ReduceLROnPlateau(monitor='val_loss', factor=0.2, patience=2, verbose=1, min_delta=0.001, min_lr=0.00001),
        tf.keras.callbacks.TensorBoard(log_dir='/kaggle/working/logs'),
        tf.keras.callbacks.EarlyStopping(patience=5, monitor='val_loss'),
        tf.keras.callbacks.ModelCheckpoint('/kaggle/working/model_for_lane_tuned_adamw_dice_batch16.keras', verbose=1, save_best_only=True)]

results = model.fit(X_train, Y_train, batch_size=16, epochs=25,steps_per_epoch=62,validation_split=0.1, callbacks=callbacks)
```

```
s = inputs

# Contraction path
c1 = tf.keras.layers.Conv2D(16, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(s)
c1 = tf.keras.layers.Dropout(0.1)(c1)
c1 = tf.keras.layers.Conv2D(16, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c1)
p1 = tf.keras.layers.MaxPooling2D((2, 2))(c1)

c2 = tf.keras.layers.Conv2D(32, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(p1)
c2 = tf.keras.layers.Dropout(0.1)(c2)
c2 = tf.keras.layers.Conv2D(32, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c2)
p2 = tf.keras.layers.MaxPooling2D((2, 2))(c2)

c3 = tf.keras.layers.Conv2D(64, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(p2)
c3 = tf.keras.layers.Dropout(0.2)(c3)
c3 = tf.keras.layers.Conv2D(64, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c3)
p3 = tf.keras.layers.MaxPooling2D((2, 2))(c3)

c4 = tf.keras.layers.Conv2D(128, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(p3)
c4 = tf.keras.layers.Dropout(0.2)(c4)
c4 = tf.keras.layers.Conv2D(128, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c4)
p4 = tf.keras.layers.MaxPooling2D(pool_size=(2, 2))(c4)

c5 = tf.keras.layers.Conv2D(256, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(p4)
c5 = tf.keras.layers.Dropout(0.3)(c5)
c5 = tf.keras.layers.Conv2D(256, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c5)

# Expansive path
u6 = tf.keras.layers.Conv2DTranspose(128, (2, 2), strides=(2, 2), padding='same')(c5)
u6 = tf.keras.layers.concatenate([u6, c4])
c6 = tf.keras.layers.Conv2D(128, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(u6)
c6 = tf.keras.layers.Dropout(0.2)(c6)
c6 = tf.keras.layers.Conv2D(128, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c6)

u7 = tf.keras.layers.Conv2DTranspose(64, (2, 2), strides=(2, 2), padding='same')(c6)
u7 = tf.keras.layers.concatenate([u7, c3])
c7 = tf.keras.layers.Conv2D(64, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(u7)
c7 = tf.keras.layers.Dropout(0.2)(c7)
c7 = tf.keras.layers.Conv2D(64, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c7)

u8 = tf.keras.layers.Conv2DTranspose(32, (2, 2), strides=(2, 2), padding='same')(c7)
u8 = tf.keras.layers.concatenate([u8, c2])
c8 = tf.keras.layers.Conv2D(32, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(u8)
c8 = tf.keras.layers.Dropout(0.1)(c8)
c8 = tf.keras.layers.Conv2D(32, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c8)

u9 = tf.keras.layers.Conv2DTranspose(16, (2, 2), strides=(2, 2), padding='same')(c8)
u9 = tf.keras.layers.concatenate([u9, c1], axis=3)
c9 = tf.keras.layers.Conv2D(16, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(u9)
c9 = tf.keras.layers.Dropout(0.1)(c9)
c9 = tf.keras.layers.Conv2D(16, (3, 3), activation='relu', kernel_initializer='he_normal', padding='same')(c9)

outputs = tf.keras.layers.Conv2D(1, (1, 1), activation='sigmoid')(c9)
```

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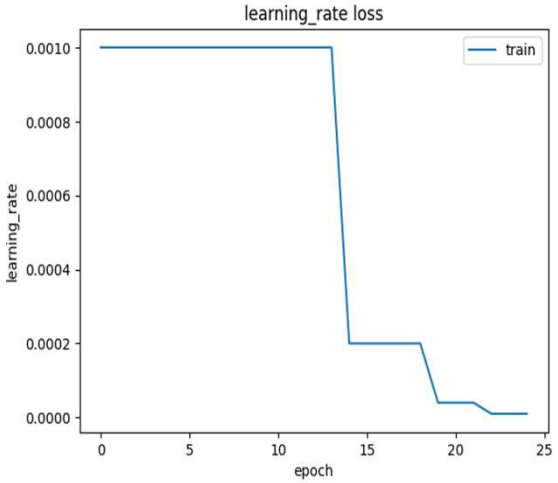
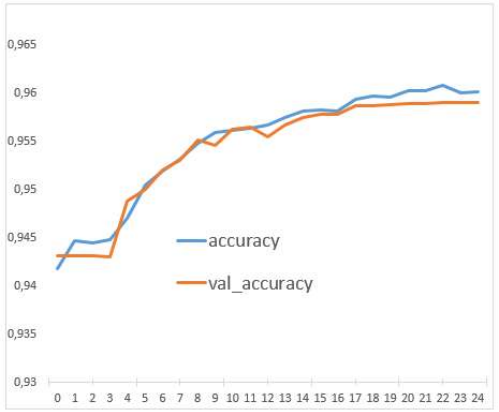
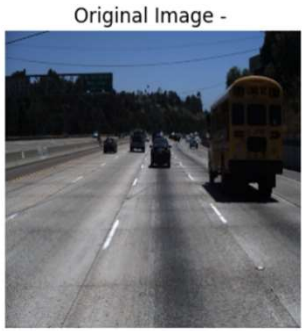
Baseline U-net results

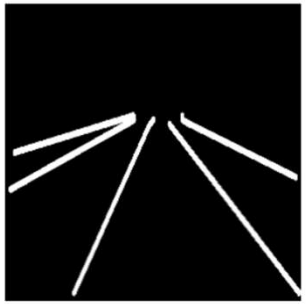
Optimization:

- **batch size: 16**
- **learning rate: 0,00001**
- **optimizer: Adam**
- **loss function: binary cross entropy**

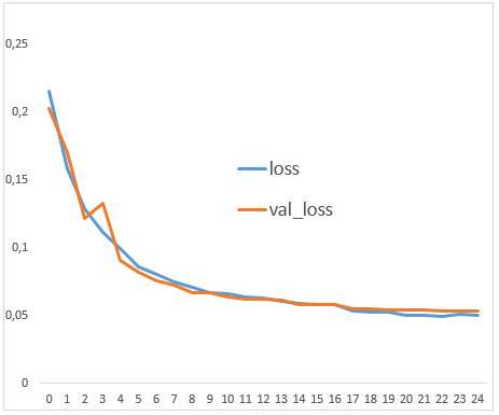








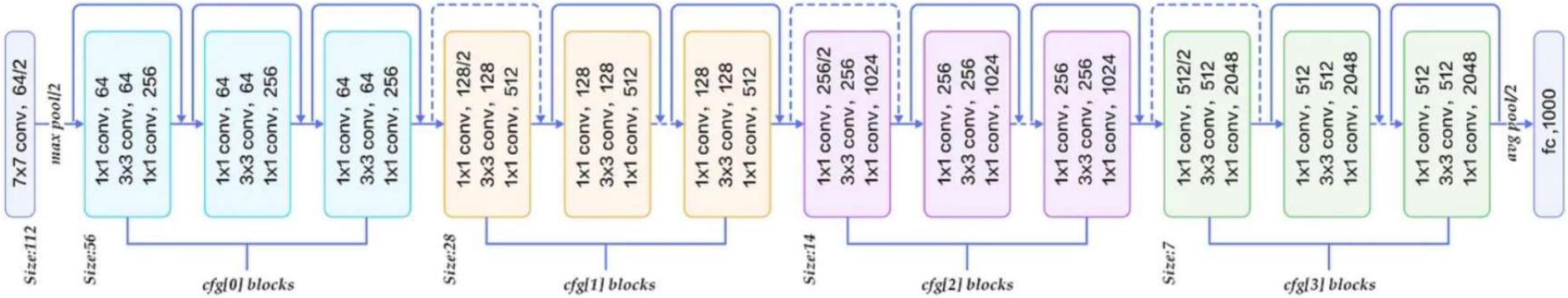

```
Precision score= 0.73
Recall Score= 0.756
Accuracy Score= 0.764
Dice Score= 0.743
Intersection Over Union= 0.591
F1 score= 0.743
```



ResNet

Further improving the model with a ResNet50 backbone.

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
		3×3 max pool, stride 2				
conv2_x	56×56	[3×3, 64; 3×3, 64] ×2	[3×3, 64; 3×3, 64] ×3	[1×1, 64; 3×3, 64; 1×1, 256] ×3	[1×1, 64; 3×3, 64; 1×1, 256] ×3	[1×1, 64; 3×3, 64; 1×1, 256] ×3
conv3_x	28×28	[3×3, 128; 3×3, 128] ×2	[3×3, 128; 3×3, 128] ×4	[1×1, 128; 3×3, 128; 1×1, 512] ×4	[1×1, 128; 3×3, 128; 1×1, 512] ×4	[1×1, 128; 3×3, 128; 1×1, 512] ×8
conv4_x	14×14	[3×3, 256; 3×3, 256] ×2	[3×3, 256; 3×3, 256] ×6	[1×1, 256; 3×3, 256; 1×1, 1024] ×6	[1×1, 256; 3×3, 256; ×1, 1024] ×23	[1×1, 256; 3×3, 256; 1×1, 1024] ×36
conv5_x	7×7	[3×3, 512; 3×3, 512] ×2	[3×3, 512; 3×3, 512] ×3	[1×1, 512; 3×3, 512; 1×1, 2048] ×3	[1×1, 512; 3×3, 512; 1×1, 2048] ×3	[1×1, 512; 3×3, 512; 1×1, 2048] ×3
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

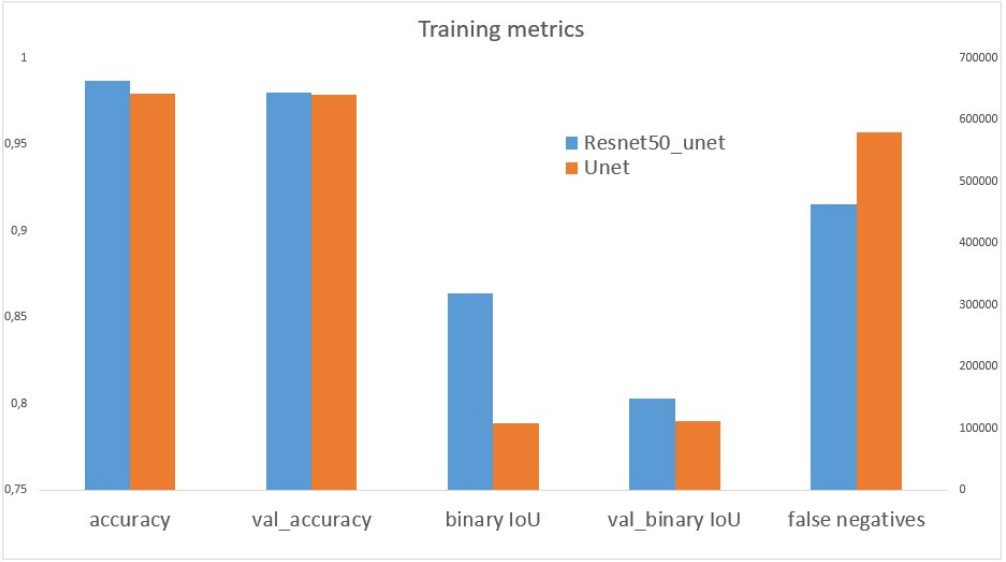


U-net + ResNet results

```python
def resnet50_encoder(input_shape):
    base_model = applications.ResNet50(include_top=False, weights='imagenet', input_shape=input_shape)
    layer_names = [
        'conv1_relu',
        'conv2_block3_out',
        'conv3_block4_out',
        'conv4_block6_out',
        'conv5_block3_out'
    ]
    layers_output = [base_model.get_layer(name).output for name in layer_names]
    return models.Model(inputs=base_model.input, outputs=layers_output)
```

```
Total params: 108,410,437 (413.55 MB)
Trainable params: 36,119,105 (137.78 MB)
Non-trainable params: 53,120 (207.50 KB)
Optimizer params: 72,238,212 (275.57 MB)
```

```
Precision score= 0.813
Recall Score= 0.794
Accuracy Score= 0.984
Dice Score= 0.803
Intersection Over Union= 0.675
F1 score= 0.803
```



Augmentation

Augmented 15% of the train pictures, then added to the base picture quantity

Methods:

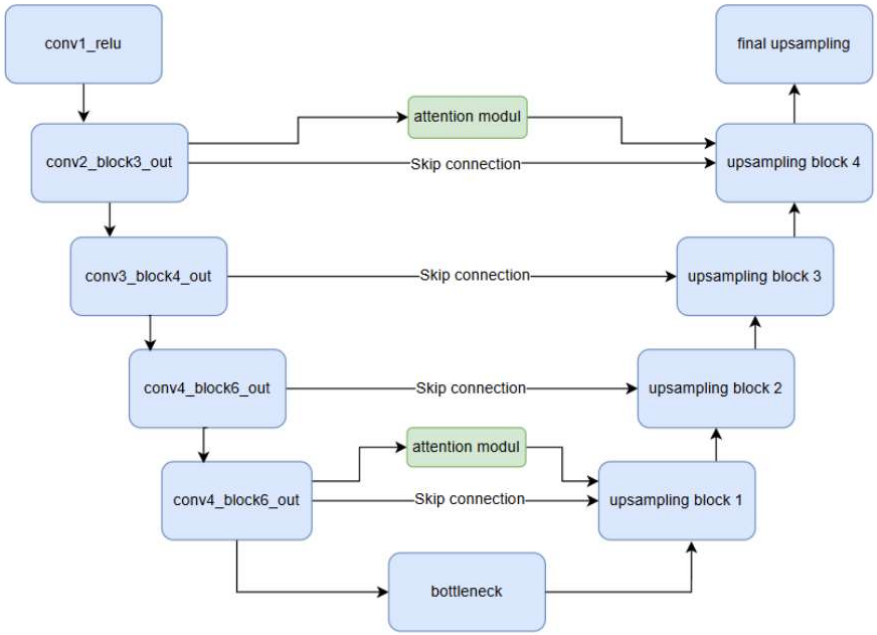
- **tf.image.flip_left_right**
- **tf.image.random_contrast(image, 0.2, 0.6)**
- **tf.image.random_brightness(image, max_delta=0.4)**

Attention modell

Spatial focus to further highlight important features

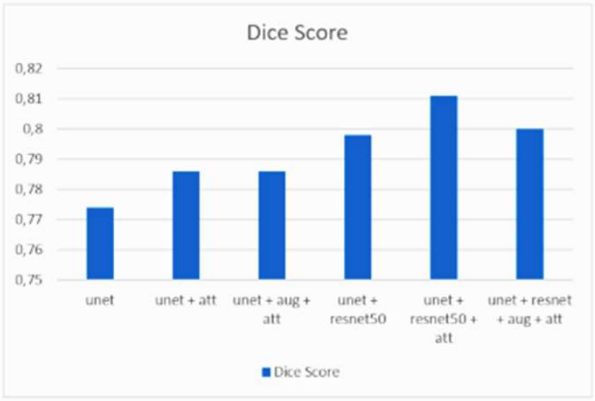
```python
def spatial_attention_module(input_tensor):

    avg_pool = tf.keras.layers.GlobalAveragePooling2D(keepdims=True)(input_tensor)
    max_pool = tf.keras.layers.GlobalMaxPooling2D(keepdims=True)(input_tensor)

    concat = tf.keras.layers.Concatenate(axis=-1)([avg_pool, max_pool])
    attention_map = tf.keras.layers.Conv2D(1, (7, 7), padding='same', activation='sigmoid')(concat)

    return input_tensor * attention_map
```

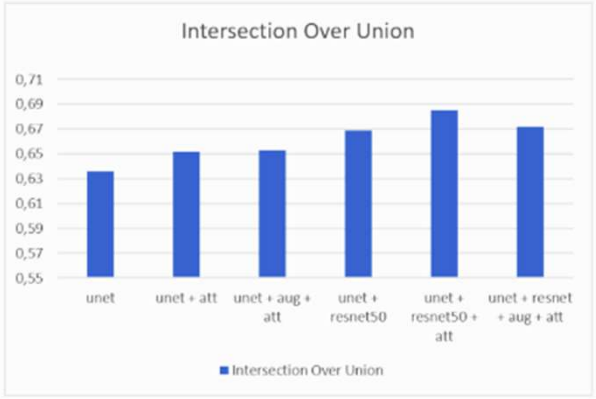
```python
x = layers.Conv2DTranspose(512, (2, 2), strides=(2, 2), padding='same')(x)
skip1 = spatial_attention_module(encoder_outputs[-2])
x = layers.concatenate([x, skip1])
x = layers.Conv2D(512, (3, 3), activation='relu', padding='same')(x)
x = layers.Dropout(dropout_rate)(x)
x = layers.Conv2D(512, (3, 3), activation='relu', padding='same')(x)
```

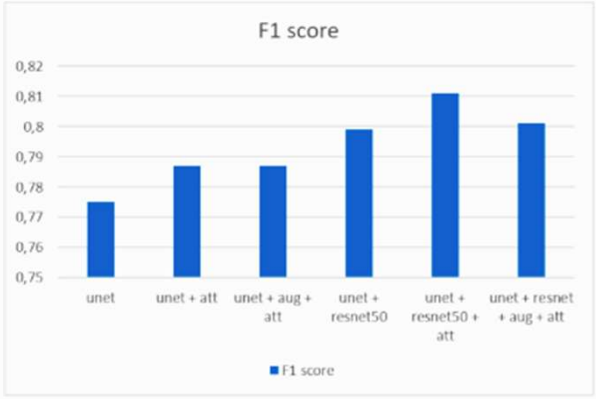


Results

Average result of 500 pcs test pictures





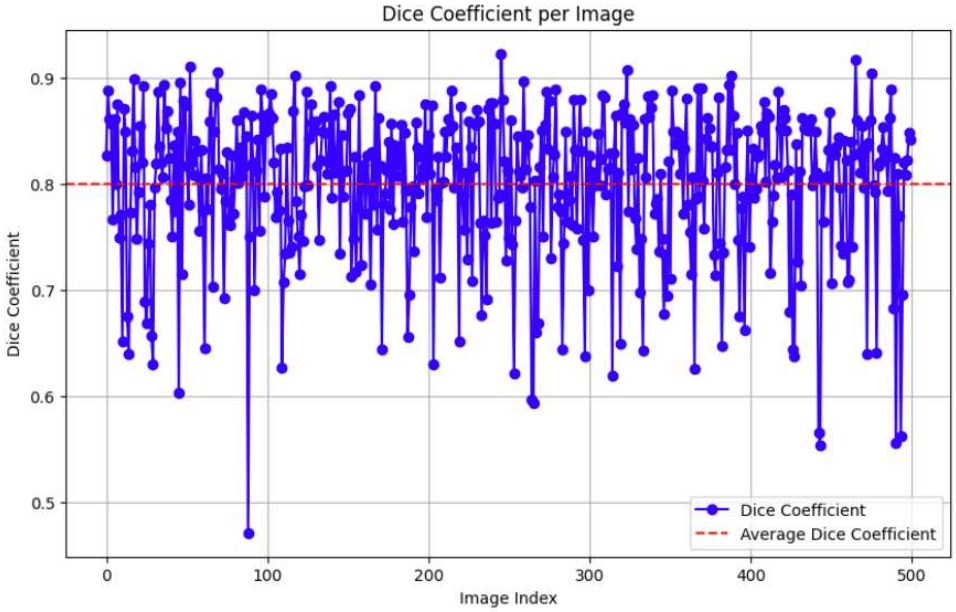


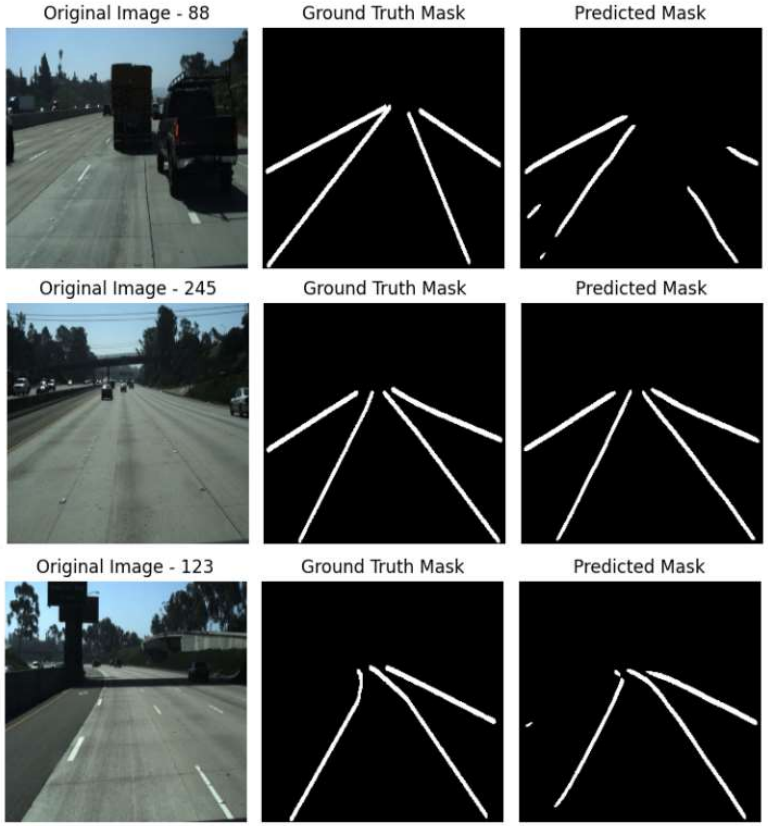
Model	Precision score	Recall Score	Accuracy Score	Dice Score	Intersection Over Union	F1 score
unet	0,793	0,758	0,981	0,774	0,636	0,775
unet + attention	0,802	0,772	0,982	0,786	0,652	0,787
unet + augmentation + attention	0,792	0,782	0,982	0,786	0,653	0,787
unet + resnet50	**0,828**	0,772	0,983	0,798	0,669	0,799
unet + resnet50 + attention	0,819	**0,804**	**0,984**	**0,811**	**0,685**	**0,811**
unet + resnet50 + augmentation + attention	0,830	0,774	0,983	0,800	0,672	0,801

Results

Distribution of 500 results (Dice score):

- **minimum (pic # 88: 0,471)**
- **maximum (pic # 245: 0,923)**
- **average (pic # 123: 0,811)**





Vehicle Detection

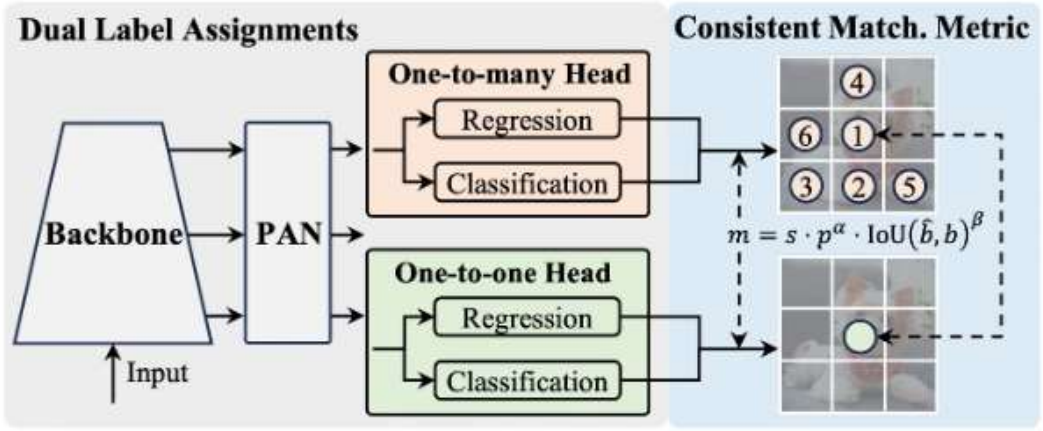
Using the YOLO model: (COCO dataset; 80 classes)

Backbone: CSPNet

One-to-many head: multiple predictions per object (training)

One-to-one head: single prediction per object (inference)

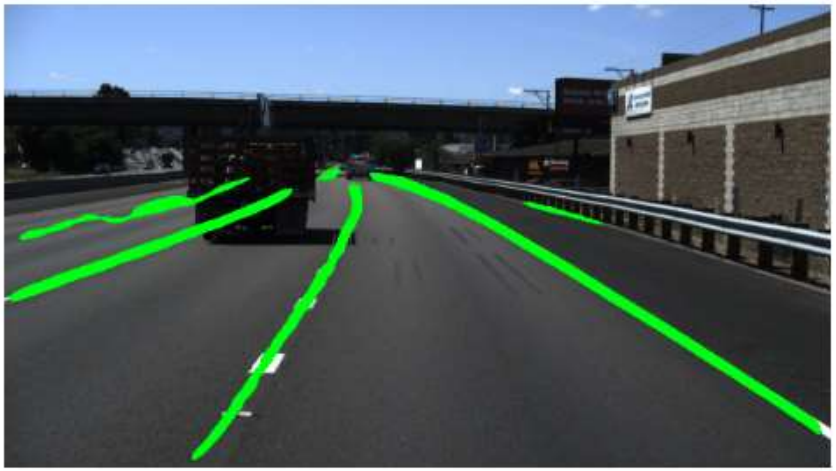
```python
from ultralytics import YOLO

# Load a pre-trained YOLOv10n model
model = YOLO("yolov10n.pt")

# Perform object detection on an image
results = model("image.jpg")

# Display the results
results[0].show()
```

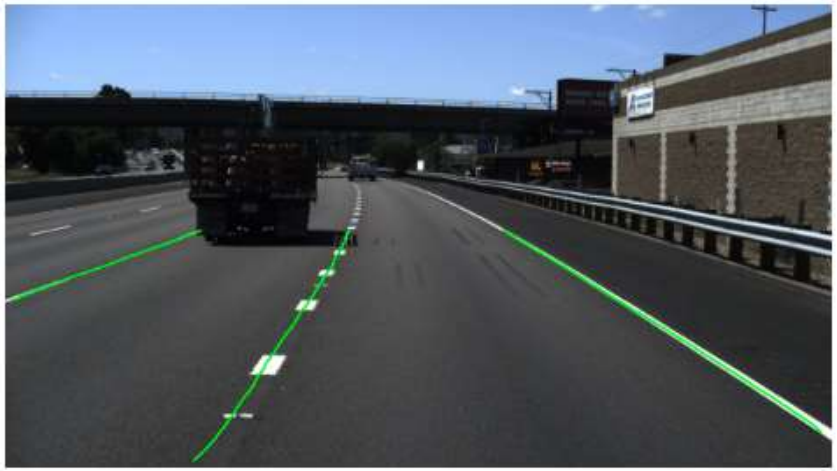


Lane Detection post-processing

Morphological improvement of output image:

- **Blur**
- **Prediction**
- **Normalize - threshold**
- **Skeletonize**
- **Region Of Interest**
- **Countour extraction -> contour length**

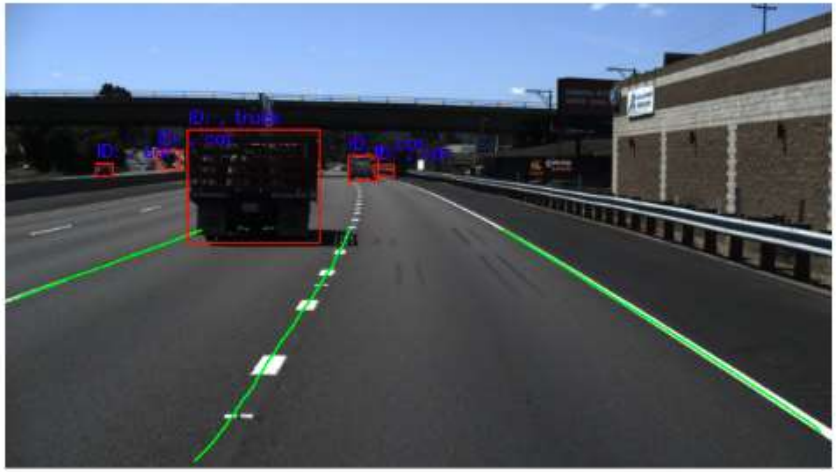




Combining Lane and Vehicle Detection

From Yolo predictions:

- **Object ID for tracking**
- **Bounding box coords**
- **Class labels**



Custom development: Ego lane

- **Identify bounding box (red rectangle)**
- **Find center X coordinate (yellow dot)**
- **Compare to image center (white dot) to determine side (left/right)**
- **Select the closest lane to the center on both sides**
- **Draw identified contours on the image**